Amino Acid Repeats:
Analysis & Implications

Ali Ahmadi Esfidi

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Abstract

Amino acid repeats (AARs) are abundant in protein sequences and play key roles in protein function and evolution. Their variation is linked to disease: expansions of polyglutamine tracts cause at least nine neurodegenerative disorders. This report reviews and implements four complementary computational strategies for detecting AARs: sliding-window Shannon entropy for low-complexity regions, a shifted self-match scan for tandem repeats, Fourier analysis for periodic patterns, and Smith–Waterman self-alignment for diverged internal repeats. A fifth method enumerates exact and BLOSUM-approximate repeated words. We give a principled, normalised periodicity score with an exact interpretation through Parseval's theorem, a vectorised self-alignment with Waterman–Eggert declumping and a shuffle-based significance estimate, and an open, tested Python implementation. On a synthetic benchmark and four real proteins (huntingtin, androgen receptor, collagen α1(I) and yeast Sup35), each method recovers the repeat class it is designed for and fails predictably on the others: entropy and tandem scanning recover the disease-associated polyQ tracts exactly, only Fourier analysis reveals the 997-residue Gly-X-Y periodicity of collagen, and self-alignment recovers the diverged oligopeptide repeats of Sup35 ($z = 6.5$). We conclude that robust AAR annotation requires combining methods, and that low-complexity masking is essential for alignment-based detection.

1 Introduction

Repetition is one of the most common architectural features of proteins. Surveys of sequence databases estimate that a substantial fraction of proteins, and a larger fraction of eukaryotic than prokaryotic proteins, contain internal repeats [1]. AARs range from runs of a single residue, such as the polyglutamine (polyQ) tract of huntingtin, through short oligopeptide units, to entire structural domains of 20–40 residues that stack into solenoids [2, 3].

AARs matter for three reasons:

- **Function and structure.** Repeated units build extended binding surfaces and fibres, as in the Gly-X-Y triple helix of collagen [4].
- **Evolution.** Repeats are mutational hotspots. Replication slippage and unequal crossing-over change copy number far faster than point mutation alters other sequence [5], which makes repeats a source of rapid variation.
- **Disease.** Expansion of CAG codon repeats beyond a threshold produces expanded polyQ tracts and causes Huntington's disease, spinal and bulbar muscular atrophy and several spinocerebellar ataxias [6–8].

Because repeats are so heterogeneous, no single algorithm detects all of them. This report (i) organises AARs into a classification framework (Section 2); (ii) describes and implements four detection strategies with precise definitions (Section 3); (iii) evaluates them on a synthetic sequence with known ground truth and on four well-characterised proteins (Section 4); and (iv) discusses their complementarity and the biological implications of repeats (Section 5).

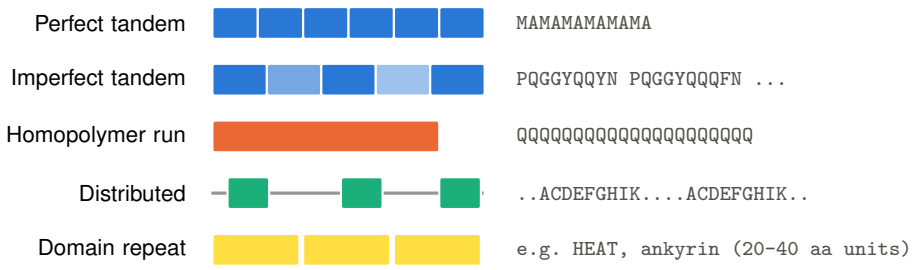


Figure 1: Classes of amino acid repeats. Shading of the imperfect units indicates divergence from the consensus. The three axes (unit similarity, spatial arrangement, pattern complexity) combine freely; a homopolymer run is the limiting case of a perfect tandem repeat with a unit length of one.

2 Classification framework

We classify repeats along three independent axes (Figure 1):

Unit similarity *Perfect* repeats have identical copies (`MAMAMA`); *imperfect* repeats tolerate substitutions or indels (`PQGGYQQYN`/`PQGGYQQQFN`).

Spatial arrangement *Tandem* copies are adjacent; *distributed* (dispersed) copies are separated by unrelated sequence.

Pattern complexity *Simple* repeats have short units (1–10 residues) and low compositional complexity; *domain* repeats have long units (typically 20–40 residues or more) that usually fold.

The axes matter because each detection strategy is sensitive to a different combination. Low compositional complexity identifies simple repeats regardless of arrangement, periodicity identifies tandem arrangements, and alignment identifies similarity regardless of arrangement.

3 Methods

Throughout, $s = s_0 s_1 \cdots s_{n-1}$ is a protein sequence over the alphabet $\mathcal{A}$ of the 20 standard amino acids. Intervals are half-open and 0-based in the equations, and 1-based and inclusive in tables and figures.

3.1 Complexity: sliding-window Shannon entropy

For a window $w = s_i \cdots s_{i+W-1}$ with residue frequencies f_a, the Shannon entropy [9]

$$H(w) = -\sum_{a \in \mathcal{A}} f_a \log_2 f_a \qquad (0 \le H \le \log_2 \min(W, 20)) \tag{1}$$

measures compositional complexity. A homopolymer has $H = 0$ and an alternating dipeptide has $H = 1$. All $n - W + 1$ window entropies are computed in $O(n|\mathcal{A}|)$ time from cumulative one-hot counts. Following SEG [10], a window with $H \le K_1$ (the *trigger*) seeds a low-complexity region (LCR). The region grows through contiguous windows with $H \le K_2$ (the *extension*), and it is the union of the residues of those windows. Because a window union overshoots the true boundary by up to $W - 1$ residues, flanking residues that occur only once within the region are trimmed. We use SEG's standard parameters $W = 12$, $K_1 = 2.2$ and $K_2 = 2.5$ bits.

3.2 Tandem repeats: shifted self-match

For a period p, define the match indicator $m_i = [s_i = s_{i+p}]$. A run of k consecutive ones means that $s_i \cdots s_{i+k+p-1}$ is p-periodic, i.e. it contains $(k + p)/p$ copies of a p-residue unit. Scanning

$p = 1, \ldots, 50$ costs $O(50\,n)$. Isolated substitutions break a run, so runs separated by short mismatch gaps are bridged. A merge is accepted only if the appended run matches the current consensus unit in phase and the merged region keeps a consensus identity of at least 80 %. This rule prevents adjacent but different homopolymers (such as $\mathtt{Q}_{21}\mathtt{P}_{11}$) from being merged. Overlapping calls are resolved by preferring longer regions and, for equal length, shorter periods, and every unit is reduced to its primitive form (PPPP→P, MAMA→MA).

3.3 Periodicity: Fourier analysis

Encoding. Fourier methods need a numeric signal. A physico-chemical scale such as hydropathy [11–14] yields a single channel and underlies classic analyses such as the hydrophobic moment [15]. However, a scale collapses residues with similar values. On the Kyte–Doolittle scale, M/A and S/T each differ by only 0.1, so $(\mathtt{MA})_n$ and $(\mathtt{ST})_n$ barely register. We therefore default to the *indicator* encoding, which has one binary channel per amino acid, $x_{a,j} = [s_j = a]$ [16]. On our synthetic sequence, the global relative power at period 2 is 1.9 with the indicator encoding and only 0.1 with hydropathy.

Global spectrum. With $X_a(f) = \sum_j (x_{a,j} - \bar{x}_a)\, e^{-2\pi i f j}$, the relative power

$$P(f) = \frac{\sum_a |X_a(f)|^2}{\sum_{a,j} (x_{a,j} - \bar{x}_a)^2} \tag{2}$$

has mean ≈ 1 over the non-zero Fourier frequencies $f = k/n$ (Parseval), so a peak value is directly an enrichment factor over a featureless spectrum.

Local periodicity score. Global spectra dilute short repeats, so we score every window. For a window of length W_p (a multiple of p), let

$$S_p = \frac{\sum_a \sum_{k=1}^{p-1} |X_a(k/p)|^2}{W_p \sum_a \sum_j (x_{a,j} - \bar{x}_a)^2}. \tag{3}$$

Because every harmonic k/p is a Fourier frequency of the window, Parseval's theorem gives $0 \le S_p \le 1$: S_p is *the fraction of the window's variance explained by period-p structure*. A perfectly p-periodic window has $S_p = 1$, and a random window has $\mathbb{E}[S_p] \approx (p-1)/W_p$, since $p-1$ of W_p frequencies are counted. We report the baseline-corrected score $E_p = (S_p - b_p)/(1 - b_p)$ with $b_p = (p-1)/W_p$, clipped to $[0, 1]$. All windows are evaluated in $O(n|\mathcal{A}|p)$ time per period using prefix sums of $x_{a,j} e^{-2\pi i k j/p}$. A p-periodic window is also periodic at every multiple of p, so the smallest period with $E_p \ge \tau$ is taken as the primitive one.

Detection. Fourier analysis proposes, string matching verifies (Algorithm 1). Periods that reach $E_p \ge \tau$ anywhere (plus their divisors) are passed to the tandem scanner of Section 3.2, and calls are kept only where the spectrum supports them. Some repeats are periodic *in pattern* but do not have identical units. Collagen's Gly-X-Y helix is the canonical example, because only every third residue is conserved. For such cases we also report *periodic regions*: maximal runs whose primitive period is constant. Period 1 (homopolymer runs) cannot be seen by the DFT, because frequency 1 aliases to frequency 0; the entropy and tandem methods cover it.

3.4 Self-comparison alignment

Diverged repeats, both tandem and distributed, are found by aligning the sequence with itself [17–19]. We fill the Smith–Waterman matrix [20] with BLOSUM62 scores σ [21] and a linear gap penalty $g = -8$:

$$H_{i,j} = \max\{0,\ H_{i-1,j-1} + \sigma(s_i, s_j),\ H_{i-1,j} + g,\ H_{i,j-1} + g\}, \tag{4}$$

Algorithm 1 Spectrum-guided tandem repeat detection

Require: sequence s, periods $p_{\min}..p_{\max}$, window W, threshold τ

$E \leftarrow$ local excess scores $E_p(i)$ for all p, i ▷ Equation (3), prefix sums

$\hat{p}(i) \leftarrow \min\{p : E_p(i) \geq \tau\}$ ▷ primitive period at i

$\mathcal{P} \leftarrow \{d : d \mid \hat{p}(i) \text{ for some } i,\ d \geq p_{\min}\}$

$R \leftarrow \textsc{TandemScan}(s, \mathcal{P})$ ▷ shifted self-match, Section 3.2

return $\{r \in R : \max_{i \in r,\ q\,:\,r.p \mid q} E_q(i) \geq \tau\}$ ▷ spectral support

restricted to the upper triangle $j > i$ to exclude the trivial diagonal and its mirror image. Each row is computed in vectorised form: with $E_j = \max\{0, H_{i-1,j-1} + \sigma, H_{i-1,j} + g\}$, the horizontal term is a running maximum, $H_{i,j} = gj + \max_{k \leq j}(E_k - gk)$, segmented at masked cells.

Declumping. The highest cell of the matrix spawns many overlapping "shadow" alignments, a problem that dominated the output of the original prototype. Following Waterman and Eggert [22], after each reported alignment its path cells are forced to zero and the matrix is recomputed from the first affected row. Alignments whose two segments both overlap a previously reported pair by more than 50 % are skipped as register-shifted duplicates.

Low-complexity masking and significance. Compositionally biased regions produce high-scoring but uninformative self-alignments (Section 4.4), so LCRs found by Section 3.1 are optionally replaced with `X` before alignment, as is standard for BLAST and RADAR. Significance is estimated with a z-score against the best self-alignment scores of 50 shuffled sequences, permuting residues only among unmasked positions. Because the null describes the *best* alignment, z-scores of lower-ranked alignments are conservative.

3.5 Repeated words: exact and approximate

The original prototype enumerated repeated substrings with a polynomial rolling hash [23] computed in 64-bit integers. For words longer than about eight residues the hash overflows, so unrelated words were silently merged and repeats were missed. We replace it with an Apriori-style growth. A word of length L can occur k times only if its length-$(L-1)$ prefix does, so only surviving start positions are extended, and the search stops at the longest repeated word. In approximate mode, every repeated word seeds a vectorised scan that accepts windows whose BLOSUM62 score reaches a fraction $\theta = 0.85$ of the word's self-score. Finally, candidates are ranked by coverage (copies × length) and accepted greedily without overlap.

3.6 Implementation and data

All methods are implemented in the Python package `aarepeats`, which depends only on NumPy [24], Matplotlib [25] and Biopython [26] (for substitution matrices). A unit-test suite of 21 tests covers known-answer cases for every detector. One script (`scripts/make_figures.py`) regenerates every figure, table and number in this report. Sequences were taken from UniProtKB/Swiss-Prot [27] (Table 1). Unless stated otherwise, parameters are: entropy $W = 12$, $K_1 = 2.2$, $K_2 = 2.5$; tandem minimum length 8, single-gap bridging, 80 % identity; Fourier $W = 12$, $\tau = 0.5$ (verified repeats) and $W = 36$, $\tau = 0.2$, minimum length 30 (periodic regions); alignment $g = -8$, minimum score 35, LCR masking on.

Table 1: Proteins analysed. Each carries a well-documented repeat of a different class.

Protein	Gene	UniProtKB	Organism	Length
Huntingtin	HTT	P42858 (v2)	*H. sapiens*	3,142
Androgen receptor	AR	P10275 (v3)	*H. sapiens*	920
Collagen α1(I) chain	COL1A1	P02452 (v6)	*H. sapiens*	1,464
Eukaryotic release factor 3 (Sup35)	SUP35	P05453 (v1)	*S. cerevisiae*	685

4 Results

4.1 Synthetic benchmark

The 66-residue synthetic sequence used in the original notebooks contains three perfect tandem repeats, three homopolymer runs and one non-repetitive spacer (Table 2, Figure 2). Entropy and tandem scanning recover every repeat element, and both correctly leave the spacer unflagged. Fourier analysis recovers all three periodic repeats and, as expected, none of the homopolymers. Self-alignment recovers everything but also flags the spacer. The spacer is caught in a long gapped alignment at a small diagonal offset that is anchored by the flanking repeats. This is the false positive that motivates low-complexity masking. Exact word enumeration misses $(\mathrm{H})_4$ and $(\mathrm{W})_5$ because they do not contain three non-overlapping copies of any word of length two or more.

4.2 Homopolymer tracts: huntingtin and the androgen receptor

In huntingtin, the tandem scanner reports a perfect $\mathtt{Q}_{21}$ tract at residues 18–38, followed immediately by $\mathtt{P}_{11}$ at 39–49. The two are kept separate by the consensus-aware bridging rule. The entropy method reports a single LCR (11–85) that spans both tracts and the proline-rich region that follows (Figure 3a). In the androgen receptor, the method reports $\mathtt{Q}_{23}$ (58–80) and $\mathtt{G}_{23}$ (451–473), which appear as the two deepest entropy minima of the whole protein (Figure 3b). These are the tracts whose expansion causes Huntington's disease and spinal and bulbar muscular atrophy (Section 5.2). Table 3 lists the longest tandem repeats in every protein.

4.3 Periodic patterns: the collagen triple helix

Collagen gives the clearest demonstration of what only spectral analysis can see. Its global spectrum has a single dominant peak at period 3.00 with relative power 102, more than five times the next-highest frequency (19; Figure 4a). The local score localises it: a period-3 region spans residues 187–1183 with mean $E_3 = 0.33$, and in that region glycine occupies 100.0 % of the

Table 2: Ground truth recovery on the synthetic sequence. ✓: at least half of the element's residues are covered by a call of that method. The spacer is a negative control, so a ✓ in that row is a false positive. Toy parameters: entropy $W = 8$, $K_1 = 1.5$, $K_2 = 2.0$; Fourier $W = 8$; tandem minimum length 4; words with at least three copies.

Element	Residues	Class	Entropy	Tandem	Fourier	Self-align.	Words
$(\mathrm{MA})_6$	1–12	perfect tandem	✓	✓	✓	✓	✓
$(\mathrm{A})_6$	26–31	homopolymer	✓	✓	✗	✓	✓
$(\mathrm{H})_4$	32–35	homopolymer	✓	✓	✗	✓	✗
$(\mathrm{ST})_5\mathrm{S}$	38–48	perfect tandem	✓	✓	✓	✓	✓
$(\mathrm{SG})_4\mathrm{S}$	49–57	perfect tandem	✓	✓	✓	✓	✓
$(\mathrm{W})_5$	58–62	homopolymer	✓	✓	✗	✓	✗
PRTEINSEQENCE	13–25	non-repeat spacer	✗	✗	✗	✓	✗

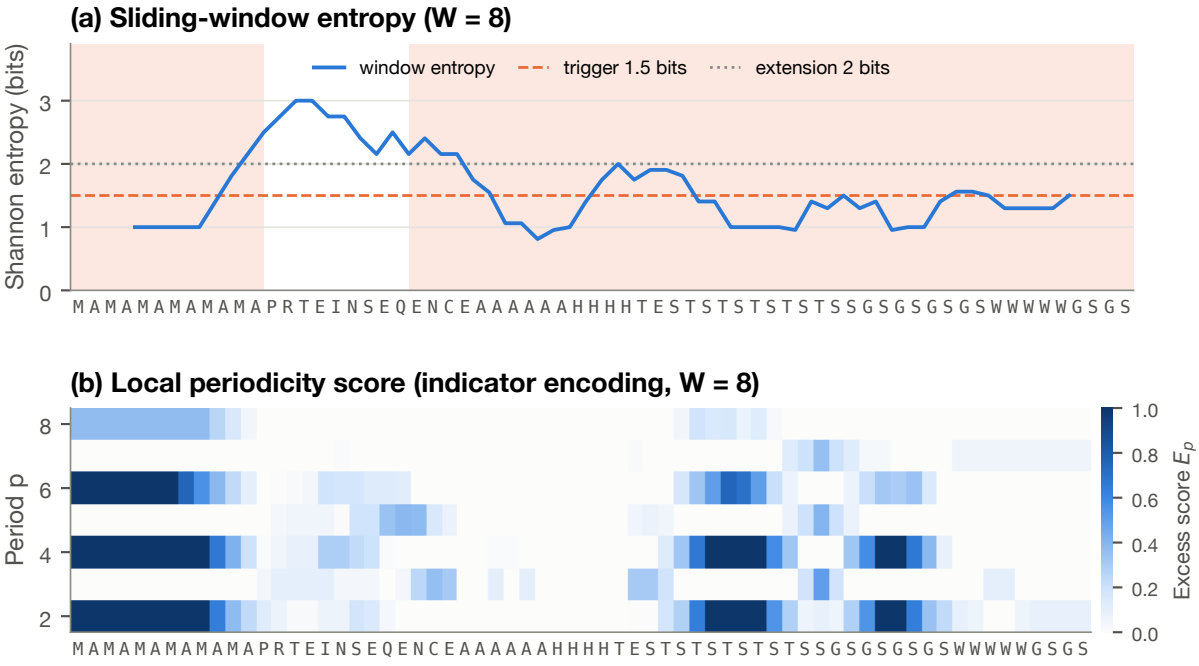


Figure 2: Synthetic sequence. (a) Window entropy with trigger and extension thresholds; shaded residues are the detected LCRs. (b) Local excess periodicity score E_p. The dipeptide repeats $\texttt{(MA)}_6$, $\texttt{(ST)}_5\texttt{S}$ and $\texttt{(SG)}_4\texttt{S}$ score $E_2 = 1$, with the expected echoes at the multiples $p = 4, 6, 8$.

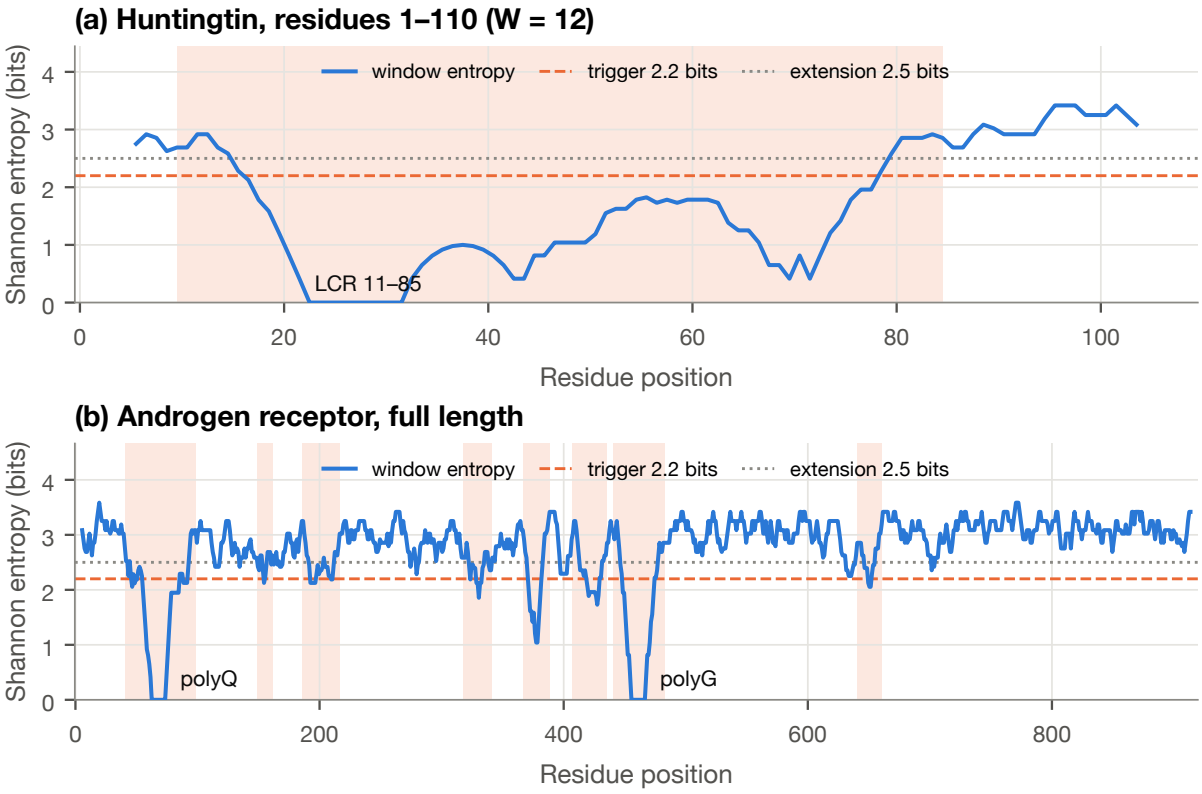


Figure 3: Sliding-window Shannon entropy ($W = 12$). Shaded regions are LCRs detected with SEG's trigger and extension thresholds. Homopolymer tracts drive the entropy to zero.

Table 3: Longest tandem repeats (up to five per protein) from the shifted self-match scan with single-gap bridging. Identity is the fraction of residues matching the consensus unit.

Protein	Residues	Unit	Period	Copies	Identity (%)
HTT	18–38	`Q`	1	21.0	100
	39–49	`P`	1	11.0	100
	63–76	`PQPQPPP`	7	2.0	86
	2649–2656	`PSSP`	4	2.0	88
AR	58–80	`Q`	1	23.0	100
	374–383	`P`	1	10.0	90
	451–473	`G`	1	23.0	100
COL1A1	138–154	`PGP`	3	5.7	94
	326–344	`GATGAAGPP`	9	2.1	89
	836–854	`GDAGPPGPA`	9	2.1	89
	884–902	`GPPGPSGNA`	9	2.1	89
	1175–1192	`GPP`	3	6.0	94
SUP35	5–12	`NQGN`	4	2.0	88
	71–90	`QQYNPQGGY`	9	2.2	100
	179–188	`TKEP`	4	2.5	100
	202–211	`TEEK`	4	2.5	90

positions in one phase of the triplet (Figure 4b). The score of about one third is itself informative. Only the glycine channel is perfectly periodic, while the X and Y positions vary, so roughly a third of the indicator variance is periodic. Collagen therefore has no long identical units: the tandem scanner fragments the helix into short pieces (Table 3), and the entropy method marks 63 % of the protein as low-complexity without revealing why. Only the periodicity analysis identifies the Gly-X-Y architecture.

4.4 Diverged repeats: self-alignment

The N-terminal domain of Sup35 contains imperfect oligopeptide repeats of the consensus `PQGGYQQYN` [28]. The tandem scanner finds only one exact two-copy stretch (`QQYNPQGGY`, residues 71–90), because the copies differ by substitutions and insertions. Self-alignment of residues 1–125 instead recovers the whole repeat array as one alignment (3–82 with 21–101; score 180, identity 52 %, $z = 6.5$) at a diagonal offset of 18 residues, about two repeat units (Figure 5a).

Low-complexity masking is not optional. Without masking, the best self-alignment of the androgen receptor spans residues 44–473 (430 residues, 31 % identity) at offset 1. In it, the polyQ, polyP and polyG tracts are chained through unrelated sequence into a single "repeat" with an apparent $z = 66$. With masking, no alignment reaches the score threshold. In collagen, masking removes the dominant helix-wide alignment, and the surviving unmasked fragments still share significant Gly-X-Y homology ($z = 7.4$). For huntingtin, whose HEAT-like domain repeats are highly diverged, the best masked alignment is not significant ($z = 1.1$). Pairwise self-alignment with a general-purpose matrix is too insensitive for such repeats, a limitation that profile- and structure-based methods address (Section 5). Figure 5b shows the masked analysis of a synthetic sequence, where the dispersed copies of `ACDEFGHIK` are recovered once the dominating dipeptide runs are masked.

4.5 Whole-protein comparison

Figure 6 and Table 4 place all methods side by side. Their outputs overlap only partly. Entropy flags compositionally simple regions, including some that are not repeats at all (for example, hydrophobic stretches in huntingtin). The tandem scanner pinpoints exact units. Fourier analysis

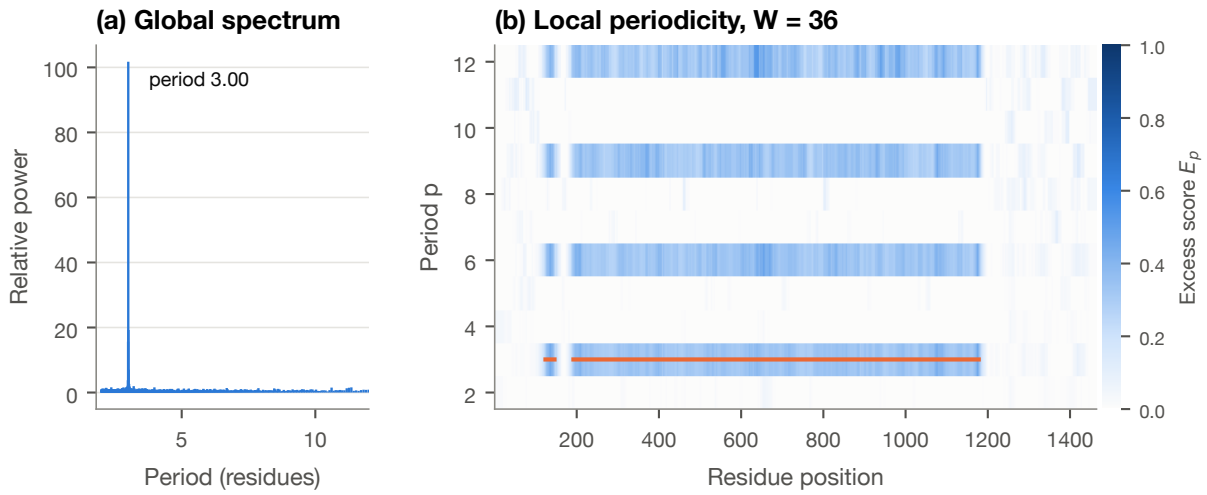


Figure 4: Collagen α1(I). (a) Global relative power spectrum (indicator encoding). (b) Local excess periodicity score; the orange line marks the detected period-3 regions. Bands at $p = 6, 9, 12$ are the expected echoes at multiples of the primitive period.

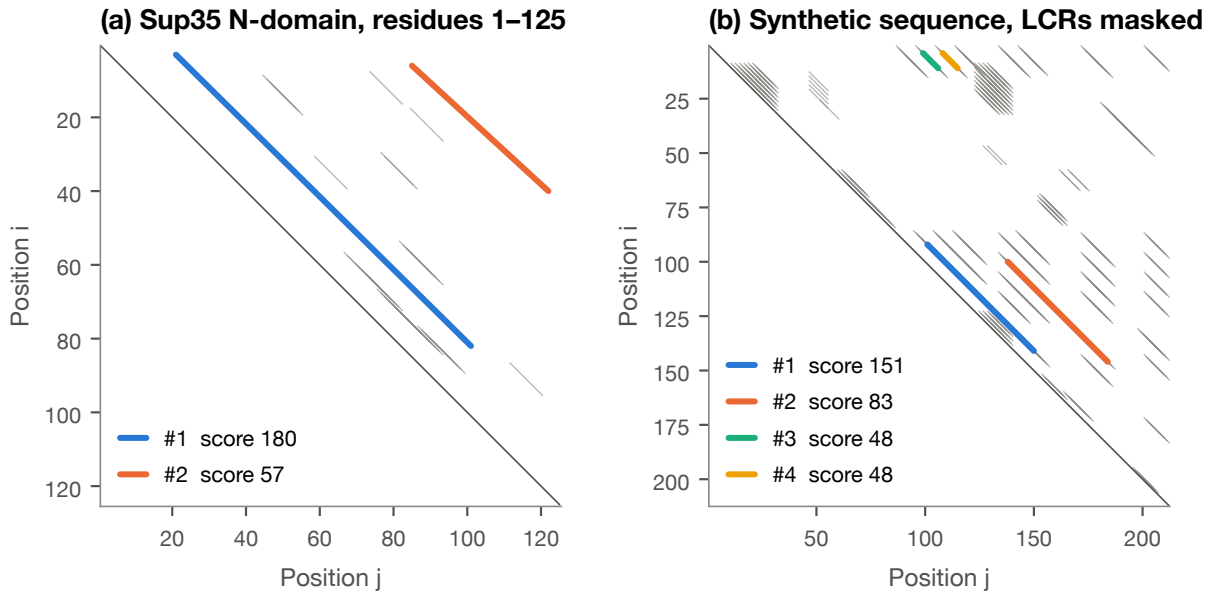


Figure 5: Self-comparison. Grey strokes: filtered dot plot (at least 6 identities in a 9-residue diagonal window). Coloured lines: non-redundant Smith–Waterman alignments after Waterman–Eggert declumping. (a) Sup35 N-domain; the main alignment links copies of the oligopeptide repeat two units apart. (b) Synthetic sequence with LCRs masked.

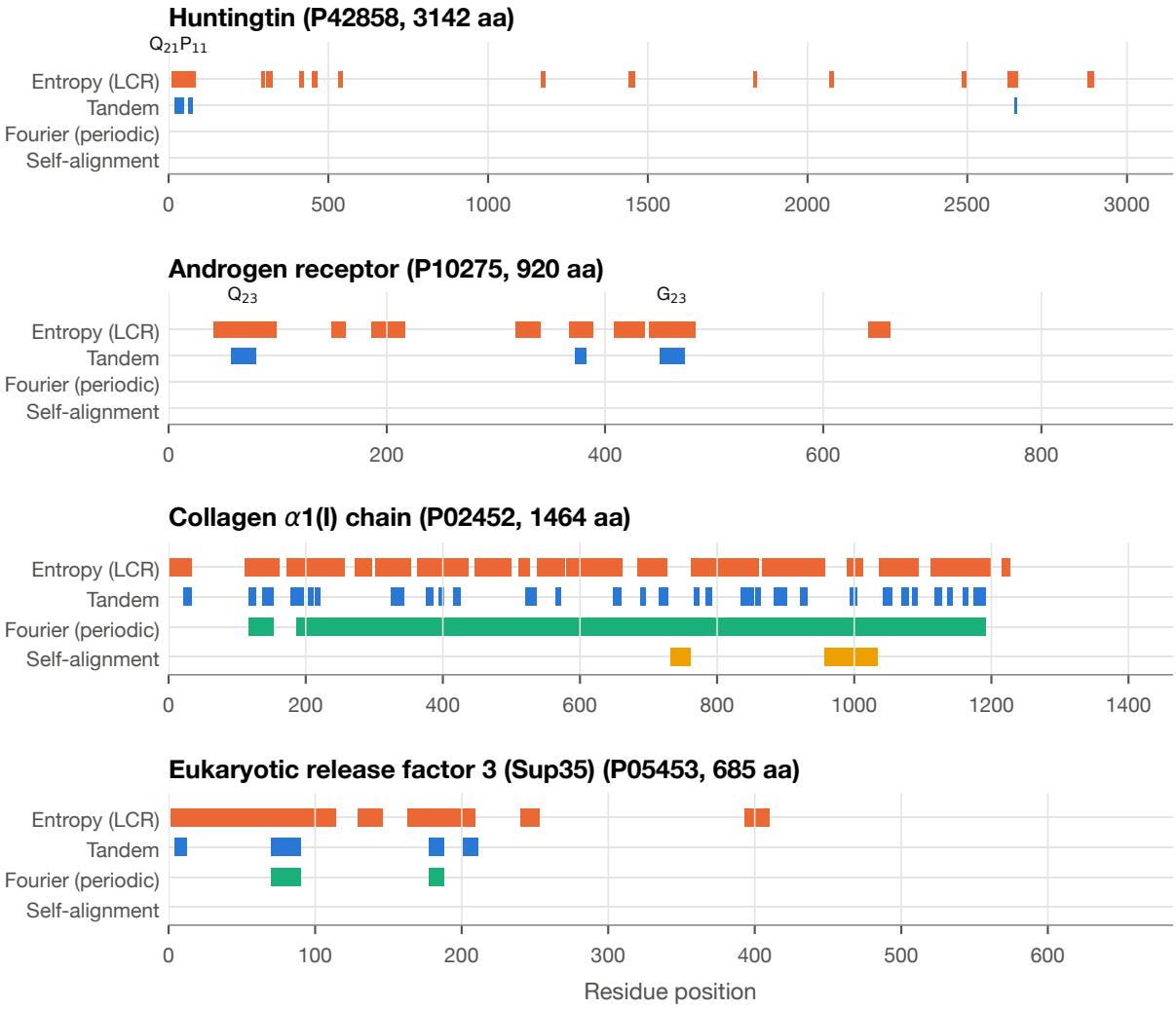


Figure 6: Repeat map of the four proteins. Each row shows the calls of one method; self-alignment shows segments of masked alignments with $z \geq 4$.

isolates periodic architecture. Self-alignment, once masked, reports only relationships between complex segments.

4.6 Computational cost

Figure 7 shows wall time on random sequences of up to 4,000 residues. Entropy and word enumeration are effectively linear and take milliseconds. The tandem and Fourier scans scale linearly with a larger constant, set by the number of periods. Self-alignment is quadratic in principle; the vectorised implementation reaches an empirical exponent of 1.8 and 0.5 s at the largest size. Every method is fast enough for proteome-scale screening of typical proteins.

Table 4: Summary per protein. Cover: percentage of residues in LCRs. Fourier verified: spectrally supported tandem repeats. Self-alignment: best masked alignment and number with $z \geq 4$ (50 shuffles).

	Entropy		Tandem	Fourier		Self-alignment (masked)	
Protein	LCRs	Cover (%)	repeats	verified	periodic regions	top score (z)	$z \geq 4$
HTT	13	8.8	4	0	–	53 (1.1)	0
AR	8	25.8	3	0	–	–	0
COL1A1	19	63.3	30	14	119–151, 187–1183 (p=3)	87 (7.4)	2
SUP35	5	30.1	4	2	–	–	0

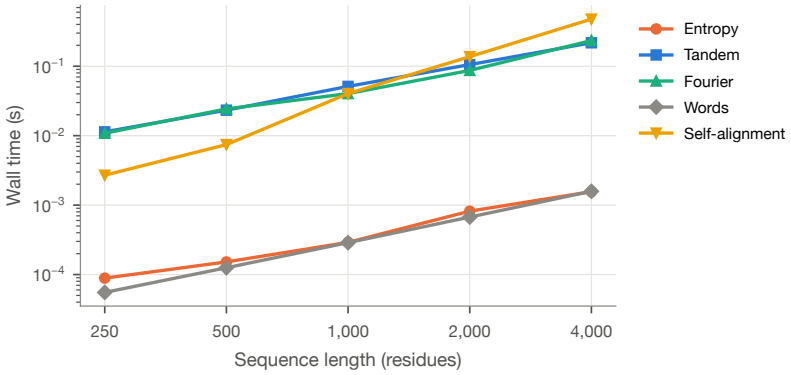


Figure 7: Wall time against sequence length on random sequences (log–log; best of three runs up to 1,000 residues). Self-alignment computes a single alignment.

5 Discussion

5.1 Complementarity of detection strategies

Table 5 summarises the strengths and blind spots that the results expose. In short, the class of a repeat (Section 2) determines which method can find it, so a robust pipeline (i) annotates LCRs by entropy, (ii) calls exact tandem units, (iii) screens for periodic architecture spectrally, and (iv) searches for diverged relationships by self-alignment after masking the LCRs from step (i).

Machine learning implementations. Statistical methods such as these remain the backbone of repeat annotation, but learned approaches are increasingly complementary. Profile-based iterative tools such as RADAR [18] and TRUST [19] already improve on pairwise alignment for

Table 5: Strengths and limitations of the detection strategies.

Method	Detects well	Misses or misleads
Entropy (SEG-style)	Homopolymers, short simple repeats, compositional bias; linear time	Diverged and domain repeats; flags non-repetitive biased regions; boundaries are only approximate
Tandem self-match	Exact or near-exact tandem units, with the unit and copy number	Distributed repeats; patterns with variable positions (Gly-X-Y); indels
Fourier	Periodic architecture without identical units; robust to substitutions	Homopolymers (period 1); non-periodic and distributed repeats; short repeats need short windows
Self-alignment	Diverged tandem and distributed repeats, with indels and significance estimates	Very diverged domain repeats; dominated by LCRs unless masked; quadratic cost
Repeated words	Exact or BLOSUM-similar dispersed motifs	Indels; spurious short words in long sequences

diverged domain repeats. Accurate structure prediction [29] makes it possible to detect repeats as recurring structural units even when their sequences have diverged beyond recognition, and classifiers trained on features such as the entropy, periodicity and alignment scores computed here are a natural next step.

5.2 Functional implications

Protein folding. Domain repeats fold into elongated solenoids (e.g. HEAT, ankyrin and leucine-rich repeats) that present large surfaces for protein–protein interaction [2, 3]. At the other extreme, the strict glycine periodicity found in collagen is a structural requirement: glycine is the only residue small enough to occupy the centre of the triple helix [4].

Evolutionary conservation patterns. Repeat copy number is unstable because slipped-strand mispairing and unequal crossing-over add or remove whole units [5]. Repeats therefore evolve faster than their flanks, and variation in their length can tune function. In Sup35, changing the number of oligopeptide repeats alters prion propagation [28].

Neurological disease associations. Expansion of polyQ tracts underlies a family of dominantly inherited neurodegenerative diseases [6]. In Huntington's disease, CAG repeats of 40 or more in *HTT* are fully penetrant, and 36–39 repeats show reduced penetrance [7]. Expansion of the androgen receptor CAG tract causes spinal and bulbar muscular atrophy [8]. Because the tract length is visible directly in sequence, the detectors here can quantify it immediately (Section 4.2).

Potential therapeutic targets. The direct link between repeat length and pathogenesis makes the expanded gene product a therapeutic target. Antisense oligonucleotides that lower huntingtin expression have been tested in patients [30], and repeat-containing proteins in general are candidates for strategies that reduce their expression or aggregation.

5.3 Limitations

Thresholds are heuristic: the SEG parameters are conventional rather than optimised, and the Fourier threshold trades sensitivity against false periods. The shuffle z-score assumes an approximately normal null for the maximum alignment score, whereas the true null is closer to an extreme-value distribution, so very small p-values should not be read literally. Finally, the evaluation uses one synthetic sequence and four proteins with well-known repeats; it illustrates the behaviour of each method but is not a benchmark of accuracy.

6 Conclusion

AARs are diverse, and so must be the methods that find them. Complexity measures, shifted self-matching, Fourier analysis and self-alignment each reveal a different facet of repetition: homopolymer tracts, exact tandem units, periodic architecture and diverged copies. On real proteins they recover the disease-associated polyQ tracts of huntingtin and the androgen receptor, the Gly-X-Y periodicity of collagen, and the diverged oligopeptide repeats of Sup35. Each method's failures are predictable from the repeat class. Combining the methods, with low-complexity masking before alignment, gives a reliable picture of a protein's repetitive architecture and of the regions where repeat variation is likely to have functional and clinical consequences.

A Corrections to the original prototype

The analysis rests on a re-implementation of the original notebooks and script. These defects were fixed:

1. **Hash overflow.** The rolling hash in `repeat detector.py` was computed in `int64` with base 256, which overflows for words longer than about eight residues, so unrelated words collided. It was replaced by exact Apriori-style word growth (Section 3.5).
2. **Mislabelled scale.** The scale labelled "Zimmerman polarity" contained Grantham's polarity values [13]. It is now labelled correctly.
3. **No mean removal.** The Fourier analysis did not subtract the signal mean, used a single global spectrum and a single physico-chemical scale. It now uses mean-centred indicator channels and the normalised local score of Equation (3).
4. **Redundant alignments.** Self-alignment reported every cell above threshold, producing dozens of overlapping copies of the same repeat, and filled both triangles of the matrix. It now uses the upper triangle, Waterman–Eggert declumping, LCR masking and shuffle z-scores.
5. **LCR boundaries.** Boundaries were derived from window centres with an asymmetric correction. Regions are now unions of qualifying windows with singleton trimming, using SEG's two thresholds.
6. **Hygiene.** The deprecated Matplotlib style `seaborn-darkgrid`, unused code, a side-effecting cache file and file names containing spaces were removed. The code was packaged, tested and documented.

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